5

6 -- Put on clothes

7 Put on blue dress

8 Put on socks and shoes

9 Wear ID card

10

11 -- Leave home

12 Take the bicycle out

13 Ride the bicycle away

Refinement is not always a sequence of actions. What the student does may depends upon the environment. How she eats breakfast depends upon how hungry she is and what is on the table; what clothes she puts on depends upon the day of the week. We can refine the behaviour which depends on environment, using conditional and iterative statements.

```
1  -- Eat breakfast
2  if hungry and idlis on the table
3      Eat idlis
4  if hungry and eggs on the table
5      Eat eggs
6  if hungry and bananas on the table
7      Eat bananas
8
8  -- Put on clothes
10 if Wednesday
11     Put on blue dress
12 else
13     Put on white dress
14 Put on socks and shoes
15 Wear the ID card
16
17 -- Leave home
18 Take the bicycle out
19 Ride the bicycle away
```

The action "Eat idlis" can be further refined as an iterative action:

```
1  -- Eat idlis
2  Put idlis on the plate
3  Add chutney
4  while idlis in plate
5      Eat a bite of idli
```

How "Get ready for school" is refined in successive levels is illustrated in Figure 7.8.

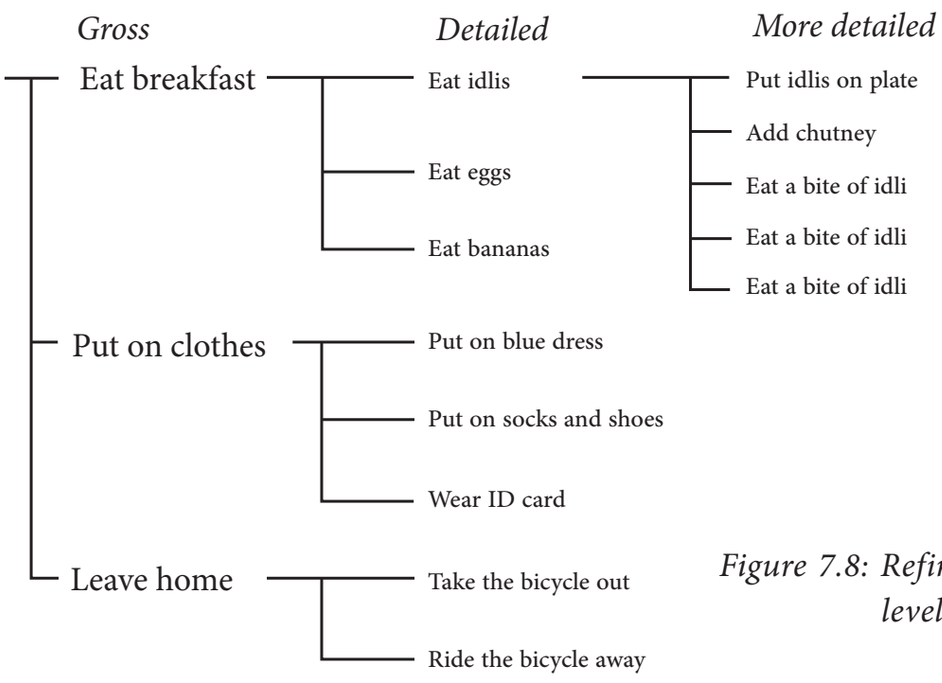

Figure 7.8: Refinement at various levels of details

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